

DTSA 5022

Statistical Learning for Data Science: Trees, SVM and Unsupervised Learning

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Data Scientist (38.1 / 100)	30.5 / 100
SPECIALIZATION Intro to Machine Learning: Supervised Learning (Statistical Learning specialization)		

30

CURRENT AVG SCORE
across 6 job roles

61

POTENTIAL SCORE
with recommended additions

+41

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Consists of the foundational framework & application of tree-based methods, support vector machines, and unsupervised learning.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	38.1	79.5	+41.4	HIGH
Applied Scientist	32.6	57.3	+24.7	MEDIUM
Machine Learning Engineer	30.9	53.3	+22.4	MEDIUM
AI Engineer	27.4	50.0	+22.6	MEDIUM
ML Engineer	27.1	58.8	+31.8	HIGH
MLOps Engineer	26.7	66.4	+39.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

5 of 6 roles

systems

Business Communication

1 of 6 roles

insights

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

GAP KEYWORDS BY ROLE

Data Scientist

38.1 → 79.5 (+41.4 pts)

data

product

insights

different

problems

Applied Scientist

32.6 → 57.3 (+24.7 pts)

computational

problems

models

data

systems

industry

world

science

Machine Learning Engineer

30.9 → 53.3 (+22.4 pts)

data

tools

science

models

model

applications

machine

systems

AI Engineer

27.4 → 50.0 (+22.6 pts)

data

models

model

systems

problems

machine

product

tools

ML Engineer

27.1 → 58.8 (+31.8 pts)

data

machine

systems

models

model

world

problems

product

MLOps Engineer

26.7 → 66.4 (+39.7 pts)

data

model

scientists

text

systems

machine

tools

models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.